PHYSICS

Time: 3 Hrs. **Max. Marks: 70**

General Instructions:

*The question paper is divided into **four** sections:*

(1) ***Section A:*** *Q. No. **1** contains **Ten multiple choice type** of questions carrying **One mark** each.*
*Q. No. **2** contains **Eight very short answer type** of questions carrying **One mark** each.*

(2) ***Section B:*** *Q. No. **3** to Q. No. **14** contain **Twelve short answer** type of questions carrying **Two marks** each. (Attempt **any Eight**).*

(3) ***Section C:*** *Q. No. **15** to Q. No. **26** contain **Twelve short answer** type of questions carrying **Three marks** each. (Attempt **any Eight**).*

(4) ***Section D:*** *Q. No. **27** to Q. No. **31** contain **Five long answer** type of questions carrying **Four marks** each. (Attempt **any Three**).*

*(5) Use of the log table is allowed. Use of calculator is **NOT** allowed.*

(6) Figures to the right indicate full marks.

(7) For multiple choice type of questions, only the first attempt will be considered for evaluation.

(8) Physical Constants:

(i) **Velocity of light in air** = 3×10^8 m/s

(ii) **Planck's constant** = 6.63×10^{-34} Js

(iii) $\pi = 3.142$

(iv) $\mu_0 = 4\pi \times 10^{-7}$ Wb/Am

(v) $1 \text{ eV} = 1.6 \times 10^{-19}$ J

(vi) $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2

(vii) **Rydberg constant** = 1.097×10^7 per meter.

SECTION – A

Q.1. Select and write the correct answers for the following multiple choice type of questions: **[10]**

i. The MI of ring of radius 'R' and mass 'M' about its central axis is _____

(A) $2\ MR^2$ (B) $\frac{MR^2}{2}$

(C) MR^2 (D) $\frac{3}{2} MR^2$

ii. For a monoatomic gas enclosed in a container at a constant temperature has degrees of freedom.

(A) 5 (B) 3

(C) 7 (D) 4

iii. The equation of the first law of thermodynamics for an isochoric process is

(A) $Q = \Delta U$ (B) $Q = 0$

(C) $Q = \Delta U + W$ (D) $Q = \Delta U - W$

iv. Cyclotron can not accelerate _____

(A) Protons (B) α^-particles

(C) Neutrons (D) Deuterons

v. In a purely inductive and capacitive circuit, power factor is, _____

(A) One (B) Zero

(C) Infinity (D) 10

vi. The Boolean equation for NOR gate is

(A) $Y = A \cdot B$ (B) $Y = A + B$

(C) $Y = \overline{A}$ (D) $Y = \overline{A + B}$

vii. When n number of capacitors having each of capacity 'C' are connected in series, their equivalent capacity is

(A) $\frac{1}{C}$ (B) nC

(C) $\frac{C}{n}$ (D) $\frac{1}{nC}$

viii. A polarising angle for an optical material is 35°. Then refractive index of the material is

(A) 0.647 (B) 0.7002

(C) 0.7265 (D) 0.7133

ix. For a streamline flow, the relative velocity between two liquid layers is 10 cm/s. The distance between them is 5 cm. The velocity gradient is

(A) 3 S ^{1} (B) 5 S ^{1}

(C) 4 S^{-1} (D) 2 S^{-1}

x. The wavelength of a stationary wave on a string is 0.75 m with frequency 64 Hz. Then, the velocity of wave is

(A) 48 m/s (B) 50 m/s

(C) 64 m/s (D) 60 m/s

Q.2. Answer the following questions: **[8]**

i. Absorption of ink by a blotting paper is an example of which phenomenon of fluids?
ii. State Kirchhoff's first law for electrical network.
iii. Define 'photoelectric work function.'
iv. Which series of spectral lines belongs to the ultraviolet region of the hydrogen spectrum?
v. Which semiconductor device converts solar energy into electrical energy?
vi. Define the term wavefront.
vii. A motorcyclist performs horizontal circular motion inside a well of inner radius 3 m. Calculate the minimum speed to perform the stunt if coefficient of static friction is 0.3. ($g = 10\ m/s^2$)
viii. Calculate the maximum velocity of a particle having amplitude of oscillation 4 cm and period 4 seconds.

SECTION – B

Attempt any EIGHT questions of the following: **[16]**

Q.3. Obtain an expression for the maximum speed of a vehicle along horizontal circular curved road.

Q.4. For a thermodynamic system,
Define: (a) Mechanical equilibrium
(b) Chemical equilibrium

Q.5. What are polar molecules and non polar molecules?

Q.6. Explain how will you convert moving coil galvanometer into voltmeter?

Q.7. Distinguish between free and forced vibrations.

Q.8. State any two postulates of Bohr's theory of hydrogen atom.

Q.9. Explain the cyclic process with the help of a neat p–V diagram.

Q.10. A star is emitting light at the wavelength 4000 Å. Determine the limit of angular resolution of a telescope having an objective diameter of 400 cm.

Q.11. Calculate the value of λ_{max} for solar radiation assuming the temperature on the surface of the sun is 5800 K. [Wien's constant = 2.897×10^{-3} m.k.]

Q.12. Calculate the radius of a water drop, so that the excess pressure inside it is 24 N/m^2. [Surface tension of water = 7.2×10^{-2} N/m]

Q.13. The susceptibility of magnetic material is χ at 27°C. At what temperature its susceptibility will be $\chi/3$?

Q.14. An alternating voltage source given by e = 140 sin (314.2 t) is connected across a pure resistor of 50 Ω. Determine:
(i) Frequency of the source.
(ii) RMS value of emf.

SECTION – C

Attempt any EIGHT questions of the following: **[24]**

Q.15. State and prove perpendicular axes theorem.

Q.16. On the basis of molecular theory, explain the phenomenon of surface tension.

Q.17. Show that all harmonics are present in the case of an air column vibrating in a pipe open at both ends.

Q.18. What is interference of light? State any four conditions for obtaining well-defined and steady interference pattern.

Q.19. With the help of a neat circuit diagram, explain the working of a half wave rectifier. Draw input and output waveforms.

Q.20. Obtain an expression for orbital magnetic moment of an electron revolving around the nucleus of an atom.

Q.21. Define:
(a) Inductive reactance
(b) Capacitive reactance
(c) Impedance

Q.22. A parallel plate capacitor filled with air has an area of 6 cm^2 and the plate separation of 2 mm. Calculate its capacity.

Q.23. Two resistances X and Y in the two gaps of a meterbridge give a null point dividing the wire in the ratio 4 : 6. If both resistances are increased by 30 Ω, the null point divides the wire in the ratio 5 : 6. Calculate the value of X and Y.

Q.24. Calculate the shortest wavelength of Paschen series and the longest wavelength of Balmer series for hydrogen atom.

Q.25. Calculate the current flowing through two long parallel wires carrying equal currents and separated by a distance of 1.35 cm experiencing a force per unit length of 4.76×10^{-2} N/m.

Q.26. An emf of 90 mV is induced in the winding of a coil, when current in a nearby coil increasing at the rate of 4.5 A/s. What is the mutual inductance of the two coils?

SECTION – D

Attempt any THREE questions of the following: **[12]**

Q.27. Derive an expression for pressure exerted by a gas on the basis of kinetic theory of gases.

Q.28. State differential equation of linear S.H.M. hence obtain an expression for acceleration, velocity and displacement of a particle performing S.H.M.

Q.29. With the help of neat diagram explain construction of a transformer.
Calculate energy required to build up current of 1A in an inductor of inductance 20 mH.

Q.30. Draw a neat schematic diagram of a photo cell.
The photoelectric work function for a metal is 4.2 eV. Find threshold wavelength.

Q.31. Obtain an expression for magnetic induction of a toroid of N no. of turns about an axis passing through its centre and perpendicular to plane.
A Carnot refrigerator operates between 150 K and 200 K, calculate its coefficient of performance.